



```
1 . do "C:\Users\fahim\AppData\Local\Temp\STD962c_000000.tmp"
```

```
2 .
```

```
3 .
```

```
4 .
```

```
5 . /* EDP -with interaction */
```

```
6 . destring EDP, replace force
   EDP already numeric; no replace
```

```
7 .
```

```
8 . xtset Number TimeNumerical
```

Panel variable: **Number** (strongly balanced)

Time variable: **TimeNumerical**, 0 to 74, but with gaps

Delta: 1 unit

```
9 . xtmixed EDP GroupNumerical##TimeNumerical Sex weight || Number:, var residuals(ar, t(TimeNumerical)) reml
```

Note: time gaps exist in the estimation data

Obtaining starting values by EM:

Performing gradient-based optimization:

Iteration 0: log restricted-likelihood = **-112.55543**

Iteration 1: log restricted-likelihood = **-112.55543**

Computing standard errors:

Mixed-effects REML regression

Group variable: **Number**

Number of obs = **52**

Number of groups = **18**

Obs per group:

min = **2**

avg = **2.9**

max = **3**

Wald chi2(10) = **21.61**

Log restricted-likelihood = **-112.55543**

Prob > chi2 = **0.0172**

EDP	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
GroupNumerical						
1	-.1489372	1.868676	-0.08	0.936	-3.811475	3.513601
2	.3575617	1.910237	0.19	0.852	-3.386435	4.101558
TimeNumerical						
30	-2.349526	1.267325	-1.85	0.064	-4.833438	.1343859
74	.4790139	1.267325	0.38	0.705	-2.004898	2.962926
GroupNumerical#TimeNumerical						
1 30	-1.657973	1.851992	-0.90	0.371	-5.28781	1.971864
1 74	-.6300144	1.792268	-0.35	0.725	-4.142796	2.882767
2 30	.3752746	1.792268	0.21	0.834	-3.137507	3.888056
2 74	.2105362	1.852734	0.11	0.910	-3.420756	3.841829
Sex	.4954183	1.418154	0.35	0.727	-2.284112	3.274949
weight	-.0147222	.1253219	-0.12	0.906	-.2603485	.2309042
_cons	7.71364	9.846228	0.78	0.433	-11.58461	27.01189

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	5.551502	2.823465	2.048773	15.04275
Residual: AR(1)				
rho	0	.	.	.
var(e)	4.818339	4.10e-43	4.818339	4.818339

LR test vs. linear model: $\chi^2(2) = 10.87$ Prob > $\chi^2 = 0.0044$

Note: LR test is conservative and provided only for reference.

10 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
EDP			
TimeNumerical@GroupNumerical			
0	2	5.71	0.0576
1	2	10.88	0.0043
2	2	4.40	0.1108
Joint	6	21.00	0.0018

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
EDP						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-2.349526	1.267325	-1.85	0.064	-4.833438	.1343859
(30 vs base) 1	-4.007499	1.350467	-2.97	0.003	-6.654365	-1.360633
(30 vs base) 2	-1.974251	1.267325	-1.56	0.119	-4.458163	.5096605
(74 vs base) 0	.4790139	1.267325	0.38	0.705	-2.004898	2.962926
(74 vs base) 1	-.1510005	1.267325	-0.12	0.905	-2.634912	2.332911
(74 vs base) 2	.6895501	1.351485	0.51	0.610	-1.959312	3.338412

11 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
EDP			
GroupNumerical@TimeNumerical			
0	2	0.07	0.9656
30	2	1.70	0.4283
74	2	0.45	0.7986
Joint	6	2.08	0.9121

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
EDP						
GroupNumerical@TimeNumerical						
(1 vs base) 0	-.1489372	1.868676	-0.08	0.936	-3.811475	3.513601
(1 vs base) 30	-1.80691	1.924341	-0.94	0.348	-5.57855	1.964729
(1 vs base) 74	-.7789516	1.868676	-0.42	0.677	-4.44149	2.883587
(2 vs base) 0	.3575617	1.910237	0.19	0.852	-3.386435	4.101558
(2 vs base) 30	.7328363	1.910237	0.38	0.701	-3.01116	4.476833
(2 vs base) 74	.5680979	1.979359	0.29	0.774	-3.311375	4.447571

```

12 .
13 . /* Repeat with natural log */
14 . generate logEDP = ln(EDP) /*root transform*/
    (2 missing values generated)

15 . xtset Number TimeNumerical

Panel variable: Number (strongly balanced)
Time variable: TimeNumerical, 0 to 74, but with gaps
Delta: 1 unit

16 . xtmixed logEDP GroupNumerical##TimeNumerical Sex weight || Number:, var residuals(ar, t(TimeNumerical)) reml
Note: time gaps exist in the estimation data

```

Obtaining starting values by EM:

Performing gradient-based optimization:

```

Iteration 0: log restricted-likelihood = -71.636857
Iteration 1: log restricted-likelihood = -71.635152
Iteration 2: log restricted-likelihood = -71.635152

```

Computing standard errors:

```

Mixed-effects REML regression
Group variable: Number
Number of obs      =      52
Number of groups   =      18
Obs per group:
    min =          2
    avg =          2.9
    max =          3
Wald chi2(10)      =      18.49
Prob > chi2        =      0.0472

Log restricted-likelihood = -71.635152

```

logEDP	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
GroupNumerical						
1	-.0799752	.6139049	-0.13	0.896	-1.283207	1.123256
2	.0259733	.6240626	0.04	0.967	-1.197167	1.249114
TimeNumerical						
30	-.4924295	.528037	-0.93	0.351	-1.527363	.542504
74	.0866662	.528037	0.16	0.870	-.9482672	1.1216
GroupNumerical#TimeNumerical						
1 30	-.7676831	.7690941	-1.00	0.318	-2.27508	.7397137
1 74	-.0359339	.7467571	-0.05	0.962	-1.499551	1.427683
2 30	-.8770152	.7467571	-1.17	0.240	-2.340632	.5866018
2 74	.0240172	.7696496	0.03	0.975	-1.484468	1.532503

Sex	.2393827	.3986793	0.60	0.548	-.5420143	1.02078
weight	-.0079181	.0354612	-0.22	0.823	-.0774207	.0615845
_cons	2.399945	2.795886	0.86	0.391	-3.079892	7.879782

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	.2856804	.2263251	.0604688	1.349675
Residual: AR(1)				
rho	5.29e-06	.	.	.
var(e)	.8364691	9.74e-29	.8364691	.8364691

LR test vs. linear model: $\chi^2(2) = 2.25$ Prob > $\chi^2 = 0.3239$

Note: LR test is conservative and provided only for reference.

17 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logEDP			
TimeNumerical@GroupNumerical			
0	2	1.40	0.4967
1	2	6.81	0.0332
2	2	9.32	0.0095
Joint	6	17.55	0.0075

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logEDP						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-.4924295	.528037	-0.93	0.351	-1.527363	.542504
(30 vs base) 1	-1.260113	.5591804	-2.25	0.024	-2.356086	-.1641391
(30 vs base) 2	-1.369445	.528037	-2.59	0.010	-2.404378	-.3345112
(74 vs base) 0	.0866662	.528037	0.16	0.870	-.9482672	1.1216
(74 vs base) 1	.0507323	.528037	0.10	0.923	-.9842012	1.085666
(74 vs base) 2	.1106834	.5599442	0.20	0.843	-.986787	1.208154

18 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logEDP			
GroupNumerical@TimeNumerical			
0	2	0.03	0.9848
30	2	2.48	0.2887
74	2	0.07	0.9669
Joint	6	2.80	0.8329

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logEDP						
GroupNumerical@TimeNumerical						
(1 vs base) 0	-.0799752	.6139049	-0.13	0.896	-1.283207	1.123256
(1 vs base) 30	-.8476583	.6400795	-1.32	0.185	-2.102191	.4068744
(1 vs base) 74	-.1159091	.6139049	-0.19	0.850	-1.319141	1.087322
(2 vs base) 0	.0259733	.6240626	0.04	0.967	-1.197167	1.249114
(2 vs base) 30	-.8510418	.6240626	-1.36	0.173	-2.074182	.3720984
(2 vs base) 74	.0499905	.657129	0.08	0.939	-1.237959	1.33794

```

19 .
20 .
21 . /* Repeat with root */
22 . generate rootEDP = sqrt(EDP) /*root transform*/
    (2 missing values generated)

23 . xtset Number TimeNumerical

    Panel variable: Number (strongly balanced)
    Time variable: TimeNumerical, 0 to 74, but with gaps
    Delta: 1 unit

24 . xtmixed rootEDP GroupNumerical##TimeNumerical Sex weight || Number:, var residuals(ar, t(TimeNumerical)) reml
    Note: time gaps exist in the estimation data

Obtaining starting values by EM:

Performing gradient-based optimization:

Iteration 0:  log restricted-likelihood = -50.801632
Iteration 1:  log restricted-likelihood = -50.80163

Computing standard errors:

Mixed-effects REML regression
Group variable: Number
Number of obs      =      52
Number of groups   =      18
Obs per group:
    min =          2
    avg =          2.9
    max =          3
Wald chi2(10)      =      29.23
Prob > chi2        =      0.0011
Log restricted-likelihood = -50.80163

```

rootEDP	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
GroupNumerical						
1	-.0618861	.4061223	-0.15	0.879	-.8578713	.734099
2	.0363488	.4148632	0.09	0.930	-.7767681	.8494657
TimeNumerical						
30	-.5236659	.2858661	-1.83	0.067	-1.083953	.0366214
74	.1007619	.2858661	0.35	0.724	-.4595254	.6610492
GroupNumerical#TimeNumerical						
1 30	-.4976518	.4176013	-1.19	0.233	-1.316135	.3208317
1 74	-.0811466	.4042757	-0.20	0.841	-.8735124	.7112193
2 30	-.1421818	.4042757	-0.35	0.725	-.9345477	.650184
2 74	.0369594	.4177844	0.09	0.930	-.7818829	.8558017
Sex	.1057491	.3029311	0.35	0.727	-.487985	.6994831
weight	-.0026028	.0267883	-0.10	0.923	-.0551068	.0499012
_cons	2.730084	2.105413	1.30	0.195	-1.396449	6.856617

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	.2448039	.1294101	.0868659	.6899016
Residual: AR(1)				
rho	-8.15e-07	.	.	.
var(e)	.2451583	2.73e-69	.2451583	.2451583

LR test vs. linear model: $\chi^2(2) = 9.13$ Prob > $\chi^2 = 0.0104$

Note: LR test is conservative and provided only for reference.

25 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
rootEDP			
TimeNumerical@GroupNumerical			
0	2	5.50	0.0639
1	2	14.72	0.0006
2	2	8.49	0.0144
Joint	6	28.73	0.0001

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
rootEDP						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-.5236659	.2858661	-1.83	0.067	-1.083953	.0366214
(30 vs base) 1	-1.021318	.3044198	-3.35	0.001	-1.61797	-.4246659
(30 vs base) 2	-.6658478	.2858661	-2.33	0.020	-1.226135	-.1055605
(74 vs base) 0	.1007619	.2858661	0.35	0.724	-.4595254	.6610492
(74 vs base) 1	.0196153	.2858661	0.07	0.945	-.5406719	.5799026
(74 vs base) 2	.1377213	.3046709	0.45	0.651	-.4594227	.7348653

26 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
rootEDP			
GroupNumerical@TimeNumerical			
0	2	0.05	0.9729
30	2	1.93	0.3803
74	2	0.26	0.8798
Joint	6	2.33	0.8869

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
rootEDP						
GroupNumerical@TimeNumerical						
(1 vs base) 0	-.0618861	.4061223	-0.15	0.879	-.8578713	.734099
(1 vs base) 30	-.559538	.4189976	-1.34	0.182	-1.380758	.2616823
(1 vs base) 74	-.1430327	.4061223	-0.35	0.725	-.9390178	.6529525
(2 vs base) 0	.0363488	.4148632	0.09	0.930	-.7767681	.8494657
(2 vs base) 30	-.1058331	.4148632	-0.26	0.799	-.91895	.7072838
(2 vs base) 74	.0733082	.4308811	0.17	0.865	-.7712033	.9178196

27 .

28 . /*EDP -without interaction */

29 . xtset Number TimeNumerical

Panel variable: **Number** (strongly balanced)

Time variable: **TimeNumerical**, 0 to 74, but with gaps

Delta: **1 unit**

```
30 . xtmixed EDP Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml
```

Performing EM optimization:

Performing gradient-based optimization:

Iteration 0: log restricted-likelihood = **-118.71558**

Iteration 1: log restricted-likelihood = **-118.71558**

Computing standard errors:

Mixed-effects REML regression	Number of obs	=	52
Group variable: Number	Number of groups	=	18
	Obs per group:		
	min	=	2
	avg	=	2.9
	max	=	3
	Wald chi2(6)	=	22.07
Log restricted-likelihood = -118.71558	Prob > chi2	=	0.0012

EDP	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Sex	.4732822	1.42262	0.33	0.739	-2.315002	3.261567
weight	-.0167412	.1255549	-0.13	0.894	-.2628243	.229342
GroupNumerical						
1	-.85962	1.567061	-0.55	0.583	-3.931004	2.211764
2	.5643047	1.621204	0.35	0.728	-2.613196	3.741805
TimeNumerical						
30	-2.722084	.7139145	-3.81	0.000	-4.121331	-1.322837
74	.3435941	.7140583	0.48	0.630	-1.055934	1.743123
_cons	8.049844	9.838565	0.82	0.413	-11.23339	27.33308

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	5.736487	2.863811	2.15627	15.26121
var(Residual)	4.407885	1.099101	2.70386	7.18582

LR test vs. linear model: chibar2(01) = 13.20 Prob >= chibar2 = **0.0001**

```
31 .
```

```
32 . contrast TimeNumerical@GroupNumerical , effect
```

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
EDP			
TimeNumerical@GroupNumerical			
0	2	21.47	0.0000
1	2	21.47	0.0000
2	2	21.47	0.0000
Joint	2	21.47	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
EDP						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-2.722084	.7139145	-3.81	0.000	-4.121331	-1.322837
(30 vs base) 1	-2.722084	.7139145	-3.81	0.000	-4.121331	-1.322837
(30 vs base) 2	-2.722084	.7139145	-3.81	0.000	-4.121331	-1.322837
(74 vs base) 0	.3435941	.7140583	0.48	0.630	-1.055934	1.743123
(74 vs base) 1	.3435941	.7140583	0.48	0.630	-1.055934	1.743123
(74 vs base) 2	.3435941	.7140583	0.48	0.630	-1.055934	1.743123

33 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
EDP			
GroupNumerical@TimeNumerical			
0	2	0.73	0.6938
30	2	0.73	0.6938
74	2	0.73	0.6938
Joint	2	0.73	0.6938

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
EDP						
GroupNumerical@TimeNumerical						
(1 vs base) 0	-.85962	1.567061	-0.55	0.583	-3.931004	2.211764
(1 vs base) 30	-.85962	1.567061	-0.55	0.583	-3.931004	2.211764
(1 vs base) 74	-.85962	1.567061	-0.55	0.583	-3.931004	2.211764
(2 vs base) 0	.5643047	1.621204	0.35	0.728	-2.613196	3.741805
(2 vs base) 30	.5643047	1.621204	0.35	0.728	-2.613196	3.741805
(2 vs base) 74	.5643047	1.621204	0.35	0.728	-2.613196	3.741805

```

34 .
35 . /* Repeat with root */
36 . xtset Number TimeNumerical

```

Panel variable: **Number** (strongly balanced)
Time variable: **TimeNumerical**, 0 to 74, but with gaps
Delta: 1 unit

```

37 . xtmixed rootEDP Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml

```

Performing EM optimization:

Performing gradient-based optimization:

Iteration 0: log restricted-likelihood = **-51.183919**
Iteration 1: log restricted-likelihood = **-51.183919**

Computing standard errors:

Mixed-effects REML regression	Number of obs	=	52
Group variable: Number	Number of groups	=	18
	Obs per group:		
	min	=	2
	avg	=	2.9
	max	=	3
	Wald chi2(6)	=	29.72
Log restricted-likelihood = -51.183919	Prob > chi2	=	0.0000

rootEDP	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Sex	.1003856	.3039668	0.33	0.741	-.4953784	.6961497
weight	-.0029757	.0268435	-0.11	0.912	-.055588	.0496367
GroupNumerical						
1	-.2415208	.3349434	-0.72	0.471	-.8979977	.4149561
2	.0022456	.3466385	0.01	0.995	-.6771533	.6816446
TimeNumerical						
30	-.7230864	.1618922	-4.47	0.000	-1.040389	-.4057835
74	.0856988	.1619283	0.53	0.597	-.2316748	.4030724
_cons	2.832823	2.103599	1.35	0.178	-1.290155	6.955801

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	.253319	.131585	.0915201	.7011632
var(Residual)	.2267561	.0565992	.139026	.369847

LR test vs. linear model: chibar2(01) = 11.06 Prob >= chibar2 = **0.0004**

38 .

39 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: asbalanced

	df	chi2	P>chi2
rootEDP			
TimeNumerical@GroupNumerical			
0	2	29.22	0.0000
1	2	29.22	0.0000
2	2	29.22	0.0000
Joint	2	29.22	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
rootEDP						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-.7230864	.1618922	-4.47	0.000	-1.040389	-.4057835
(30 vs base) 1	-.7230864	.1618922	-4.47	0.000	-1.040389	-.4057835
(30 vs base) 2	-.7230864	.1618922	-4.47	0.000	-1.040389	-.4057835
(74 vs base) 0	.0856988	.1619283	0.53	0.597	-.2316748	.4030724
(74 vs base) 1	.0856988	.1619283	0.53	0.597	-.2316748	.4030724
(74 vs base) 2	.0856988	.1619283	0.53	0.597	-.2316748	.4030724

40 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: asbalanced

	df	chi2	P>chi2
rootEDP			
GroupNumerical@TimeNumerical			
0	2	0.65	0.7228
30	2	0.65	0.7228
74	2	0.65	0.7228
Joint	2	0.65	0.7228

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
rootEDP						
GroupNumerical@TimeNumerical						
(1 vs base) 0	-.2415208	.3349434	-0.72	0.471	-.8979977	.4149561
(1 vs base) 30	-.2415208	.3349434	-0.72	0.471	-.8979977	.4149561
(1 vs base) 74	-.2415208	.3349434	-0.72	0.471	-.8979977	.4149561
(2 vs base) 0	.0022456	.3466385	0.01	0.995	-.6771533	.6816446
(2 vs base) 30	.0022456	.3466385	0.01	0.995	-.6771533	.6816446
(2 vs base) 74	.0022456	.3466385	0.01	0.995	-.6771533	.6816446

```

41 .
42 . /* Repeat with log */
43 . xtset Number TimeNumerical

```

Panel variable: **Number** (strongly balanced)
Time variable: **TimeNumerical**, 0 to 74, but with gaps
Delta: **1 unit**

```

44 . xtmixed logEDP Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml

```

Performing EM optimization:

Performing gradient-based optimization:

Iteration 0: log restricted-likelihood = **-74.716263**
Iteration 1: log restricted-likelihood = **-74.715377**
Iteration 2: log restricted-likelihood = **-74.715377**

Computing standard errors:

Mixed-effects REML regression	Number of obs	=	52
Group variable: Number	Number of groups	=	18
	Obs per group:		
	min	=	2
	avg	=	2.9
	max	=	3
	Wald chi2(6)	=	17.24
Log restricted-likelihood = -74.715377	Prob > chi2	=	0.0085

logEDP	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Sex	.2365784	.3966175	0.60	0.551	-.5407776	1.013934
weight	-.0074157	.0352112	-0.21	0.833	-.0764284	.0615969
GroupNumerical						
1	-.3366756	.4382868	-0.77	0.442	-1.195702	.5223508
2	-.2686952	.4549794	-0.59	0.555	-1.160438	.6230481
TimeNumerical						
30	-1.030244	.3019541	-3.41	0.001	-1.622063	-.4384248
74	.0741164	.3020712	0.25	0.806	-.5179324	.6661652
_cons	2.544647	2.760912	0.92	0.357	-2.866642	7.955936

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	.2952065	.2330886	.062811	1.387446
var(Residual)	.7914493	.1973228	.4855154	1.290159

LR test vs. linear model: chibar2(01) = 2.73 Prob >= chibar2 = **0.0491**

45 .

46 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logEDP			
TimeNumerical@GroupNumerical			
0	2	16.29	0.0003
1	2	16.29	0.0003
2	2	16.29	0.0003
Joint	2	16.29	0.0003

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logEDP						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-1.030244	.3019541	-3.41	0.001	-1.622063	-.4384248
(30 vs base) 1	-1.030244	.3019541	-3.41	0.001	-1.622063	-.4384248
(30 vs base) 2	-1.030244	.3019541	-3.41	0.001	-1.622063	-.4384248
(74 vs base) 0	.0741164	.3020712	0.25	0.806	-.5179324	.6661652
(74 vs base) 1	.0741164	.3020712	0.25	0.806	-.5179324	.6661652
(74 vs base) 2	.0741164	.3020712	0.25	0.806	-.5179324	.6661652

47 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logEDP			
GroupNumerical@TimeNumerical			
0	2	0.67	0.7147
30	2	0.67	0.7147
74	2	0.67	0.7147
Joint	2	0.67	0.7147

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logEDP						
GroupNumerical@TimeNumerical						
(1 vs base) 0	-.3366756	.4382868	-0.77	0.442	-1.195702	.5223508
(1 vs base) 30	-.3366756	.4382868	-0.77	0.442	-1.195702	.5223508
(1 vs base) 74	-.3366756	.4382868	-0.77	0.442	-1.195702	.5223508
(2 vs base) 0	-.2686952	.4549794	-0.59	0.555	-1.160438	.6230481
(2 vs base) 30	-.2686952	.4549794	-0.59	0.555	-1.160438	.6230481
(2 vs base) 74	-.2686952	.4549794	-0.59	0.555	-1.160438	.6230481

```

48 .
49 .
50 . /* EDV -with interaction */
51 . destring EDV, replace force
    EDV already numeric; no replace

```

```

52 .
53 . xtset Number TimeNumerical

```

```

Panel variable: Number (strongly balanced)
Time variable: TimeNumerical, 0 to 74, but with gaps
Delta: 1 unit

```

```

54 . xtmixed EDV GroupNumerical##TimeNumerical Sex weight || Number:, var reml

```

Performing EM optimization:

Performing gradient-based optimization:

```

Iteration 0: log restricted-likelihood = -184.81645
Iteration 1: log restricted-likelihood = -184.81645

```

Computing standard errors:

```

Mixed-effects REML regression
Group variable: Number
Number of obs   =      52
Number of groups =      18
Obs per group:
    min =      2
    avg =     2.9
    max =      3
Wald chi2(10)   =    151.53
Prob > chi2     =     0.0000
Log restricted-likelihood = -184.81645

```

EDV	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
GroupNumerical						
1	12.9741	12.65089	1.03	0.305	-11.8212	37.7694
2	-14.70988	12.97919	-1.13	0.257	-40.14863	10.72886
TimeNumerical						
30	-42.67399	6.64226	-6.42	0.000	-55.69258	-29.6554
74	-3.591627	6.64226	-0.54	0.589	-16.61022	9.426964
GroupNumerical#TimeNumerical						
1 30	-1.373056	9.721907	-0.14	0.888	-20.42764	17.68153
1 74	13.1838	9.393575	1.40	0.160	-5.227267	31.59487
2 30	8.499128	9.393575	0.90	0.366	-9.91194	26.9102
2 74	-.9149418	9.724075	-0.09	0.925	-19.97378	18.14389
Sex	-2.867398	10.41215	-0.28	0.783	-23.27483	17.54004
weight	1.444271	.9172919	1.57	0.115	-.3535882	3.24213
_cons	-5.279423	71.96217	-0.07	0.942	-146.3227	135.7638

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	342.0968	154.8407	140.8887	830.6573
var(Residual)	132.3589	35.53949	78.19846	224.0309

LR test vs. linear model: $\text{chibar2}(01) = 20.19$ Prob >= chibar2 = **0.0000**

55 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
EDV			
TimeNumerical@GroupNumerical			
0	2	50.79	0.0000
1	2	62.69	0.0000
2	2	30.49	0.0000
Joint	6	143.99	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
EDV						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-42.67399	6.64226	-6.42	0.000	-55.69258	-29.6554
(30 vs base) 1	-44.04705	7.099004	-6.20	0.000	-57.96084	-30.13326
(30 vs base) 2	-34.17486	6.64226	-5.15	0.000	-47.19346	-21.15627
(74 vs base) 0	-3.591627	6.64226	-0.54	0.589	-16.61022	9.426964
(74 vs base) 1	9.592174	6.64226	1.44	0.149	-3.426417	22.61077
(74 vs base) 2	-4.506569	7.101972	-0.63	0.526	-18.42618	9.413041

56 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
EDV			
GroupNumerical@TimeNumerical			
0	2	4.28	0.1178
30	2	1.78	0.4104
74	2	9.72	0.0077
Joint	6	12.11	0.0595

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
EDV						
GroupNumerical@TimeNumerical						
(1 vs base) 0	12.9741	12.65089	1.03	0.305	-11.8212	37.7694
(1 vs base) 30	11.60104	12.88937	0.90	0.368	-13.66165	36.86374
(1 vs base) 74	26.1579	12.65089	2.07	0.039	1.362606	50.9532
(2 vs base) 0	-14.70988	12.97919	-1.13	0.257	-40.14863	10.72886
(2 vs base) 30	-6.210754	12.97919	-0.48	0.632	-31.6495	19.22799
(2 vs base) 74	-15.62482	13.27273	-1.18	0.239	-41.6389	10.38925

```

57 .
58 . /* Repeat with root */
59 . generate rootEDV = sqrt(EDV) /*root transform*/
    (2 missing values generated)

60 . xtset Number TimeNumerical

Panel variable: Number (strongly balanced)
Time variable: TimeNumerical, 0 to 74, but with gaps
Delta: 1 unit

61 . xtmixed rootEDV GroupNumerical##TimeNumerical Sex weight || Number:, var residuals(ar, t(TimeNumerical)) reml
Note: time gaps exist in the estimation data

```

Obtaining starting values by EM:

Performing gradient-based optimization:

```

Iteration 0: log restricted-likelihood = -58.32298
Iteration 1: log restricted-likelihood = -58.32298

```

Computing standard errors:

```

Mixed-effects REML regression
Group variable: Number
Number of obs      =      52
Number of groups   =      18
Obs per group:
    min =          2
    avg =          2.9
    max =          3
Wald chi2(10)      =     210.45
Prob > chi2        =     0.0000

Log restricted-likelihood = -58.32298

```

rootEDV	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
GroupNumerical						
1	.5596182	.6150583	0.91	0.363	-.6458739	1.76511
2	-.7287305	.6316285	-1.15	0.249	-1.9667	.5092387
TimeNumerical						
30	-2.318619	.2925542	-7.93	0.000	-2.892015	-1.745223
74	-.1660482	.2925542	-0.57	0.570	-.7394439	.4073475
GroupNumerical#TimeNumerical						
1 30	.0969699	.4283519	0.23	0.821	-.7425844	.9365243
1 74	.5248059	.4137341	1.27	0.205	-.2860981	1.33571
2 30	.4394196	.4137341	1.06	0.288	-.3714843	1.250324
2 74	-.0164858	.4284288	-0.04	0.969	-.8561908	.8232192
Sex	-.126023	.5162792	-0.24	0.807	-1.137912	.8858657

weight	.0616052	.0454509	1.36	0.175	-.027477	.1506874
_cons	5.544095	3.564477	1.56	0.120	-1.442151	12.53034

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	.8641823	.377746	.3668879	2.035529
Residual: AR(1)				
rho	1.05e-08	.	.	.
var(e)	.2567639	4.05e-51	.2567639	.2567639

LR test vs. linear model: $\chi^2(2) = 24.79$ Prob > $\chi^2 = 0.0000$

Note: LR test is conservative and provided only for reference.

62 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
rootEDV			
TimeNumerical@GroupNumerical			
0	2	78.18	0.0000
1	2	76.86	0.0000
2	2	48.75	0.0000
Joint	6	203.81	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
rootEDV						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-2.318619	.2925542	-7.93	0.000	-2.892015	-1.745223
(30 vs base) 1	-2.221649	.3128856	-7.10	0.000	-2.834894	-1.608405
(30 vs base) 2	-1.879199	.2925542	-6.42	0.000	-2.452595	-1.305804
(74 vs base) 0	-.1660482	.2925542	-0.57	0.570	-.7394439	.4073475
(74 vs base) 1	.3587577	.2925542	1.23	0.220	-.214638	.9321533
(74 vs base) 2	-.182534	.3129909	-0.58	0.560	-.7959848	.4309168

63 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
rootEDV			
GroupNumerical@TimeNumerical			
0	2	3.91	0.1414
30	2	2.18	0.3361
74	2	7.79	0.0203
Joint	6	9.84	0.1313

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
rootEDV						
GroupNumerical@TimeNumerical						
(1 vs base) 0	.5596182	.6150583	0.91	0.363	-.6458739	1.76511
(1 vs base) 30	.6565882	.6246946	1.05	0.293	-.5677907	1.880967
(1 vs base) 74	1.084424	.6150583	1.76	0.078	-.1210681	2.289916
(2 vs base) 0	-.7287305	.6316285	-1.15	0.249	-1.9667	.5092387
(2 vs base) 30	-.2893109	.6316285	-0.46	0.647	-1.52728	.9486583
(2 vs base) 74	-.7452163	.6434649	-1.16	0.247	-2.006384	.5159518

```

64 .
65 .
66 . /* Repeat with natural log */
67 . generate logEDV = ln(EDV) /*root transform*/
    (2 missing values generated)

68 . xtset Number TimeNumerical

    Panel variable: Number (strongly balanced)
    Time variable: TimeNumerical, 0 to 74, but with gaps
    Delta: 1 unit

69 . xtmixed logEDV GroupNumerical##TimeNumerical Sex weight || Number:, var residuals(ar, t(TimeNumerical)) reml
    Note: time gaps exist in the estimation data

Obtaining starting values by EM:

Performing gradient-based optimization:

Iteration 0:   log restricted-likelihood = 9.2708576
Iteration 1:   log restricted-likelihood = 9.2708576

Computing standard errors:

Mixed-effects REML regression
Group variable: Number
Number of obs      =      52
Number of groups   =      18
Obs per group:
    min =          2
    avg =          2.9
    max =          3
Wald chi2(10)      =     271.27
Prob > chi2        =     0.0000
Log restricted-likelihood = 9.2708576

```

logEDV	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
GroupNumerical						
1	.0968984	.1234265	0.79	0.432	-.145013	.3388099
2	-.1460423	.126822	-1.15	0.250	-.3946089	.1025243
TimeNumerical						
30	-.5086024	.0548897	-9.27	0.000	-.6161842	-.4010206
74	-.0306037	.0548897	-0.56	0.577	-.1381856	.0769781
GroupNumerical#TimeNumerical						
1 30	.0526868	.0803846	0.66	0.512	-.1048641	.2102376
1 74	.0832537	.0776258	1.07	0.283	-.06889	.2353974
2 30	.0902588	.0776258	1.16	0.245	-.0618849	.2424024
2 74	.0020274	.080397	0.03	0.980	-.1555479	.1596027
Sex	-.0222593	.1047497	-0.21	0.832	-.2275649	.1830464
weight	.0105898	.0092181	1.15	0.251	-.0074773	.0286569
_cons	3.845111	.7227962	5.32	0.000	2.428456	5.261765

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	.0360901	.	.	.
Residual: AR(1)				
rho	1.37e-06	.	.	.
var(e)	.0090386	1.18e-34	.0090386	.0090386

LR test vs. linear model: $\chi^2(2) = 28.01$ Prob > $\chi^2 = 0.0000$

Note: LR test is conservative and provided only for reference.

70 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logEDV			
TimeNumerical@GroupNumerical			
0	2	108.00	0.0000
1	2	87.19	0.0000
2	2	70.18	0.0000
Joint	6	265.40	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logEDV						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-.5086024	.0548897	-9.27	0.000	-.6161842	-.4010206
(30 vs base) 1	-.4559156	.0587265	-7.76	0.000	-.5710174	-.3408138
(30 vs base) 2	-.4183436	.0548897	-7.62	0.000	-.5259254	-.3107618
(74 vs base) 0	-.0306037	.0548897	-0.56	0.577	-.1381856	.0769781
(74 vs base) 1	.05265	.0548897	0.96	0.337	-.0549318	.1602318
(74 vs base) 2	-.0285763	.0587436	-0.49	0.627	-.1437116	.0865589

71 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logEDV			
GroupNumerical@TimeNumerical			
0	2	3.47	0.1766
30	2	2.61	0.2706
74	2	6.01	0.0497
Joint	6	7.68	0.2624

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logEDV						
GroupNumerical@TimeNumerical						
(1 vs base) 0	.0968984	.1234265	0.79	0.432	-.145013	.3388099
(1 vs base) 30	.1495852	.1251287	1.20	0.232	-.0956625	.3948329
(1 vs base) 74	.1801521	.1234265	1.46	0.144	-.0617593	.4220636
(2 vs base) 0	-.1460423	.126822	-1.15	0.250	-.3946089	.1025243
(2 vs base) 30	-.0557836	.126822	-0.44	0.660	-.3043502	.192783
(2 vs base) 74	-.1440149	.1289104	-1.12	0.264	-.3966747	.1086448

72 .

73 .

74 .

75 . /*EDV -without interaction */

76 . xtset Number TimeNumerical

Panel variable: **Number** (strongly balanced)

Time variable: **TimeNumerical**, 0 to 74, but with gaps

Delta: **1 unit**

```
77 . xtmixed EDV Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml
```

Performing EM optimization:

Performing gradient-based optimization:

Iteration 0: log restricted-likelihood = **-200.20283**

Iteration 1: log restricted-likelihood = **-200.20283**

Computing standard errors:

Mixed-effects REML regression	Number of obs	=	52
Group variable: Number	Number of groups	=	18
	Obs per group:		
	min	=	2
	avg	=	2.9
	max	=	3
	Wald chi2(6)	=	136.21
Log restricted-likelihood = -200.20283	Prob > chi2	=	0.0000

EDV	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Sex	-3.109739	10.55107	-0.29	0.768	-23.78945	17.56998
weight	1.390437	.9292261	1.50	0.135	-.4308129	3.211687
GroupNumerical						
1	17.28811	11.60864	1.49	0.136	-5.464407	40.04062
2	-11.56281	11.99524	-0.96	0.335	-35.07304	11.94743
TimeNumerical						
30	-39.84079	4.043016	-9.85	0.000	-47.76496	-31.91663
74	.9285864	4.04352	0.23	0.818	-6.996567	8.85374
_cons	-3.39689	72.8011	-0.05	0.963	-146.0844	139.2906

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	349.5748	158.8538	143.4621	851.8109
var(Residual)	141.1779	35.43715	86.31899	230.9018

LR test vs. linear model: chibar2(01) = 21.32 Prob >= chibar2 = **0.0000**

```
78 .
```

```
79 . contrast TimeNumerical@GroupNumerical , effect
```

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
EDV			
TimeNumerical@GroupNumerical			
0	2	128.87	0.0000
1	2	128.87	0.0000
2	2	128.87	0.0000
Joint	2	128.87	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
EDV						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-39.84079	4.043016	-9.85	0.000	-47.76496	-31.91663
(30 vs base) 1	-39.84079	4.043016	-9.85	0.000	-47.76496	-31.91663
(30 vs base) 2	-39.84079	4.043016	-9.85	0.000	-47.76496	-31.91663
(74 vs base) 0	.9285864	4.04352	0.23	0.818	-6.996567	8.85374
(74 vs base) 1	.9285864	4.04352	0.23	0.818	-6.996567	8.85374
(74 vs base) 2	.9285864	4.04352	0.23	0.818	-6.996567	8.85374

80 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
EDV			
GroupNumerical@TimeNumerical			
0	2	5.49	0.0644
30	2	5.49	0.0644
74	2	5.49	0.0644
Joint	2	5.49	0.0644

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
EDV						
GroupNumerical@TimeNumerical						
(1 vs base) 0	17.28811	11.60864	1.49	0.136	-5.464407	40.04062
(1 vs base) 30	17.28811	11.60864	1.49	0.136	-5.464407	40.04062
(1 vs base) 74	17.28811	11.60864	1.49	0.136	-5.464407	40.04062
(2 vs base) 0	-11.56281	11.99524	-0.96	0.335	-35.07304	11.94743
(2 vs base) 30	-11.56281	11.99524	-0.96	0.335	-35.07304	11.94743
(2 vs base) 74	-11.56281	11.99524	-0.96	0.335	-35.07304	11.94743

```

81 .
82 . /* Repeat with root */
83 . xtset Number TimeNumerical

```

Panel variable: **Number** (strongly balanced)
Time variable: **TimeNumerical**, 0 to 74, but with gaps
Delta: 1 unit

```
84 . xtmixed rootEDV Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml
```

Performing EM optimization:

Performing gradient-based optimization:

Iteration 0: log restricted-likelihood = **-60.46673**
Iteration 1: log restricted-likelihood = **-60.46673**

Computing standard errors:

Mixed-effects REML regression	Number of obs	=	52
Group variable: Number	Number of groups	=	18
	Obs per group:		
	min	=	2
	avg	=	2.9
	max	=	3
	Wald chi2(6)	=	201.82
Log restricted-likelihood = -60.46673	Prob > chi2	=	0.0000

rootEDV	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Sex	-.134202	.5228214	-0.26	0.797	-1.158913	.8905091
weight	.0595217	.0460098	1.29	0.196	-.0306559	.1496992
GroupNumerical						
1	.7779907	.5749815	1.35	0.176	-.3489523	1.904934
2	-.5627151	.5938758	-0.95	0.343	-1.72669	.60126
TimeNumerical						
30	-2.125583	.1740263	-12.21	0.000	-2.466669	-1.784498
74	.021137	.174043	0.12	0.903	-.3199809	.362255
_cons	5.584679	3.604456	1.55	0.121	-1.479925	12.64928

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	.8869644	.3879242	.3763772	2.090206
var(Residual)	.2614431	.0655629	.1599258	.4274012

LR test vs. linear model: chibar2(01) = 27.41 Prob >= chibar2 = **0.0000**

85 .

86 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
rootEDV			
TimeNumerical@GroupNumerical			
0	2	195.35	0.0000
1	2	195.35	0.0000
2	2	195.35	0.0000
Joint	2	195.35	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
rootEDV						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-2.125583	.1740263	-12.21	0.000	-2.466669	-1.784498
(30 vs base) 1	-2.125583	.1740263	-12.21	0.000	-2.466669	-1.784498
(30 vs base) 2	-2.125583	.1740263	-12.21	0.000	-2.466669	-1.784498
(74 vs base) 0	.021137	.174043	0.12	0.903	-.3199809	.362255
(74 vs base) 1	.021137	.174043	0.12	0.903	-.3199809	.362255
(74 vs base) 2	.021137	.174043	0.12	0.903	-.3199809	.362255

87 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
rootEDV			
GroupNumerical@TimeNumerical			
0	2	4.81	0.0905
30	2	4.81	0.0905
74	2	4.81	0.0905
Joint	2	4.81	0.0905

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
rootEDV						
GroupNumerical@TimeNumerical						
(1 vs base) 0	.7779907	.5749815	1.35	0.176	-.3489523	1.904934
(1 vs base) 30	.7779907	.5749815	1.35	0.176	-.3489523	1.904934
(1 vs base) 74	.7779907	.5749815	1.35	0.176	-.3489523	1.904934
(2 vs base) 0	-.5627151	.5938758	-0.95	0.343	-1.72669	.60126
(2 vs base) 30	-.5627151	.5938758	-0.95	0.343	-1.72669	.60126
(2 vs base) 74	-.5627151	.5938758	-0.95	0.343	-1.72669	.60126


```

88 .
89 .
90 . /* Repeat with log */
91 . xtset Number TimeNumerical

```

Panel variable: **Number** (strongly balanced)
Time variable: **TimeNumerical**, 0 to 74, but with gaps
Delta: 1 unit

```

92 . xtmixed logEDV Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml

```

Performing EM optimization:

Performing gradient-based optimization:

Iteration 0: log restricted-likelihood = **14.588228**
Iteration 1: log restricted-likelihood = **14.588228**

Computing standard errors:

Mixed-effects REML regression	Number of obs	=	52
Group variable: Number	Number of groups	=	18
	Obs per group:		
	min	=	2
	avg	=	2.9
	max	=	3
	Wald chi2(6)	=	275.40
Log restricted-likelihood = 14.588228	Prob > chi2	=	0.0000

logEDV	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Sex	-.0231379	.1058993	-0.22	0.827	-.2306967	.184421
weight	.0102902	.0093155	1.10	0.269	-.0079678	.0285482
GroupNumerical						
1	.1429473	.1164365	1.23	0.220	-.085264	.3711586
2	-.1114122	.1202337	-0.93	0.354	-.3470659	.1242414
TimeNumerical						
30	-.4597691	.0319037	-14.41	0.000	-.5222991	-.3972391
74	.0006427	.0319062	0.02	0.984	-.0618923	.0631777
_cons	3.842379	.7297589	5.27	0.000	2.412078	5.27268

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	.0370431	.0158696	.0159973	.0857766
var(Residual)	.0087844	.0022017	.0053749	.0143566

LR test vs. linear model: chibar2(01) = 32.08 Prob >= chibar2 = **0.0000**

93 .

94 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logEDV			
TimeNumerical@GroupNumerical			
0	2	269.66	0.0000
1	2	269.66	0.0000
2	2	269.66	0.0000
Joint	2	269.66	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logEDV						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-.4597691	.0319037	-14.41	0.000	-.5222991	-.3972391
(30 vs base) 1	-.4597691	.0319037	-14.41	0.000	-.5222991	-.3972391
(30 vs base) 2	-.4597691	.0319037	-14.41	0.000	-.5222991	-.3972391
(74 vs base) 0	.0006427	.0319062	0.02	0.984	-.0618923	.0631777
(74 vs base) 1	.0006427	.0319062	0.02	0.984	-.0618923	.0631777
(74 vs base) 2	.0006427	.0319062	0.02	0.984	-.0618923	.0631777

95 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logEDV			
GroupNumerical@TimeNumerical			
0	2	4.20	0.1226
30	2	4.20	0.1226
74	2	4.20	0.1226
Joint	2	4.20	0.1226

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logEDV						
GroupNumerical@TimeNumerical						
(1 vs base) 0	.1429473	.1164365	1.23	0.220	-.085264	.3711586
(1 vs base) 30	.1429473	.1164365	1.23	0.220	-.085264	.3711586
(1 vs base) 74	.1429473	.1164365	1.23	0.220	-.085264	.3711586
(2 vs base) 0	-.1114122	.1202337	-0.93	0.354	-.3470659	.1242414
(2 vs base) 30	-.1114122	.1202337	-0.93	0.354	-.3470659	.1242414
(2 vs base) 74	-.1114122	.1202337	-0.93	0.354	-.3470659	.1242414

96 .

97 .

98 .

end of do-file

99 .